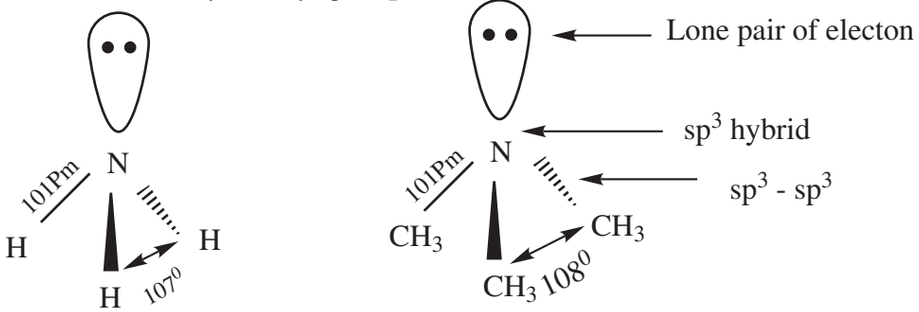
13.2.2 Structure of amines

Like, ammonia, nitrogen atom of amines is trivalent and carries a lone pair of electron and sp^3 hybridised, out of the four sp^3 hybridised orbitals of nitrogen, three sp^3 orbitals overlap with orbitals of hydrogen (or) alkyl groups of carbon, the fourth sp^3 orbital contains a lone pair of electron. Hence, amines posses pyramidal geometry. Due to presence of lone pair of electron C- N- H (or) C- N- C bond angle is less than the normal tetrahedral bond angle 109.5^0. For example, the C- N- C bond angle of trimethylamine is 108^0 which is lower than tetrahedral angle and higher than the H- N- H bond angle of 107°. This increase is due to the repulsion between the bulky methyl groups.



13.2.3 General methods of preparation Amines

Aliphatic and aromatic amines are prepared by the following methods.

1) From nitro compounds

Reduction of Nitro compounds using H_2 / Ni (or) Sn / HCl or Pd/H_2 gives primary amines.

$$CH_3CH_2 - NO_2 \xrightarrow[Fe\ /\ HCl,\ 6\ [H]]{3H_2\ /\ Ni\ (or)} CH_3CH_2 - NH_2 + 2\ H_2O$$

Nitroethane — ethanamine

$$C_6H_5 - NO_2 \xrightarrow[(or)\ Sn\ /\ HCl]{3H_2\ /\ Pt,\ 680\ K} C_6H_5 - NH_2 + 2\ H_2O$$

Nitrobenzene — Aniline

2) From nitriles

a) Reduction of alkyl or aryl cyanides with H_2/Ni (or) $LiAlH_4$ (or) Na / C_2H_5OH gives primary amines. The reduction reaction in which Na / C_2H_5OH is used as a reducing agent is called mendius reaction

$$CH_3 - CN \xrightarrow[4\ [\ H\ ]]{Na(Hg)\ /\ C_2H_5OH} CH_3CH_2 - NH_2$$

ethanenitrile — ethanamine

b) Reduction of isocyanides with sodium amalgam / C_2H_5OH gives secondary amines

$$CH_3 - NC \xrightarrow[4\ [\ H]]{Na(Hg)\ /\ C_2H_5OH} CH_3 - NH - CH_3$$

Methyl isocyanide — N-methylmethanamine

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